

Esercizio.

Suppose M c.t.m. of ZFC, $M \models k \geq 2^\omega \wedge k \in \text{Reg}$. Let $P := (P_k)^M, B := RO(P)^M$. Then $\forall G$ generic $\omega_1^M = \omega_1^{M[G]}$.

Dimostrazione.

By contradiction, assume that for some G generic we have $M[G] \models \omega_1 > (\omega_1)^M$. This means that there exists in $M[G]$ some r such that $M[G] \models r : \omega_1^M \rightarrow \omega$ injective. Let $\dot{r}$ be a name for r . Then $[[\dot{r} : \omega_1^M \rightarrow \omega]] = b \in G$. Now consider for every $n \in \omega$ $D_n := \{q \leq b : M \models \exists n < \omega [[\dot{r}(\check{n}) = \check{m}]] \geq q\}$. We claim that $\forall n < \omega D_n$ is open, dense under b . It is obvious that it is open, since it is downward closed by definition. We need to check the density: let $0 < c \leq b$. Let H be generic, with $c \in H$. Then $b \in H$, so that $M[H] \models \dot{r}_H : \omega_1^M \rightarrow \omega$ injection. In particular, for some $m < \omega$, it must be $M[H] \models \dot{r}_H(n) = m$. Equivalently $[[\dot{r}(\check{n}) = \check{m}]] \in H$. Then $[[\dot{r}(\check{n}) = \check{m}]] \wedge c \in H \cap D_n$. Now, since P is ω -complete, distributive we get that B is ω -distributive. This means in particular that $D_{\dot{r}} := \bigcap_{n < \omega} D_n$ is open, dense under b . Since G is generic, $b \in G$ and $D_{\dot{r}}$ is a dense of M , we get that $G \cap D_{\dot{r}} \neq \emptyset$. So let $q \in G \cap D_{\dot{r}}$. We claim $q \Vdash [[\dot{r} = \check{s}]]$ for some $s \in M$. In fact, consider $s := \{\langle n, m \rangle : q \Vdash [[\dot{r}(\check{n}) = \check{m}]]\}$. Let H generic with $q \in H$. Then $M[G] \models \dot{r}_H(n) = m = \check{s}_H(n)$. Since this holds for every n we get that $M[H] \models \dot{r}_H = \check{s}_H$. Since H arbitrary we also get $q \Vdash [[\dot{r} = \check{s}]]$. Since $q \in G$ we finally have $M[G] \models \dot{r}_G = s$. But this means that the function witnessing $|\omega_1^M| = \omega$ already existed in M , a contradiction.

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